Aerospace Technology

Essential answers to the "To think at home" questions

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Short version of the answers to the *To think at home* questions: the idea, the result and how to rebuild it, with the same numbering as the full notes. The index lists every answer and each entry is a link.

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UNIT 9

Essential answers

This PDF keeps only what you need to answer each *To think at home* question: the idea, the result and a way to rebuild it if you forget the formula. The numbering is the same as in the full notes: **9.1.***n* is slide question n and **9.2.***n* is question n of the professor's list. Every formula has its symbols defined right below it. For derivations, worked examples and the extra figures, open the full notes (`Apuntes_Tecnologia_Aeroespacial.pdf`).

1 Essential answers to the questions written on the slides

1.1 Slide question 1: navigation aids

Working principles of the barometric altimeter, radio altimeter, NDB, VOR, ILS, GNSS and GBAS. Differences and uses.

MUST KNOW

- **Barometric altimeter**: measures static pressure and converts it to altitude with the ISA. Height above a *pressure level*, not above the ground. Used everywhere; flight levels use one common reference.
- **Radio altimeter**: radar pointing down, $h = c\,\Delta t/2$. Height *above the ground*, accurate, short range. Final approach, flare, autoland, terrain warning.
- **NDB**: ground beacon; the ADF antenna points to it. Gives only the *direction*; poor accuracy. Old aid.
- **VOR**: compares two signal phases; gives the *radial* (bearing from the station), about 1°. With DME (distance) it fixes the position. Airways.
- **ILS**: two beams (localizer and glide slope) compare tone strengths; gives lateral and vertical *guidance* to one runway. Precision approach.
- **GNSS**: signal delays from satellites, at least **4** satellites (3 coordinates plus the receiver clock error). Global, few metres, no integrity guarantee.
- **GBAS**: a ground station at a known position computes the pseudorange errors and broadcasts corrections by VHF. Sub-metre accuracy and integrity, so it supports precision approach.

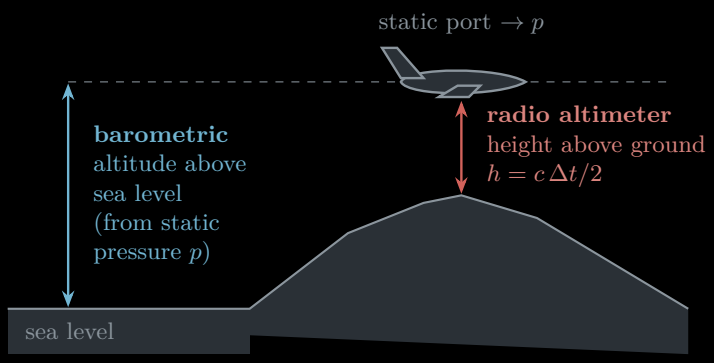


Figure 9.1. Barometric altimeter (altitude above a pressure level) against radio altimeter (height above whatever is below).

HOW TO REMEMBER

Ask two questions of each aid: *what does it give?* (height, direction, distance, guidance, position) and *where is the reference?* (pressure level, ground, ground beacon, runway, satellites). The two altimeters give height; NDB and VOR give direction from a beacon; ILS gives a runway path; GNSS gives position anywhere; GBAS is GNSS made precise near an airport.

1.2 Slide question 2: Magnus effect

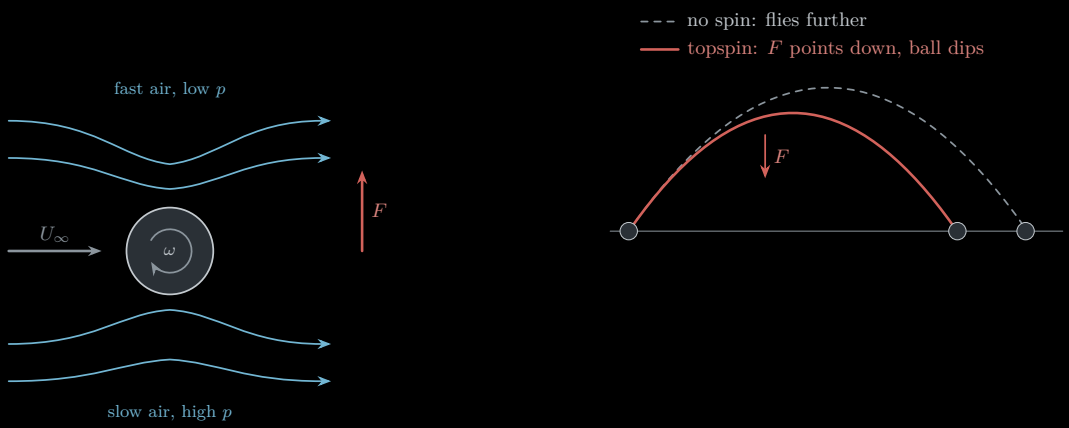
Where do we see the Magnus effect?

MUST KNOW

A spinning body in an airstream feels a force **perpendicular to the wind**. The side whose surface moves *with* the stream has faster air (lower pressure); the force points to that side.

- **Where**: ball sports (topspin dips, backspin floats, banana kicks), Flettner rotor ships, spinning projectiles.
- **Why it matters**: lift is a consequence of **circulation**, $L' = \rho V \Gamma$ (Kutta–Joukowski). A cylinder gets its circulation from the spin; an airfoil gets it from the Kutta condition at the trailing edge.

Here L' is the lift per unit span, ρ the air density, V the free-stream speed and Γ the circulation.



Magnus effect
top surface moves with the stream: faster air,
lower pressure; force ⊥ stream, towards the low p side

Consequence
with topspin the force adds to weight,
so the ball lands sooner

Figure 9.2. Spinning cylinder: faster air on the side that moves with the stream, slower on the other, net force F towards the low-pressure side.

1.3 Slide question 3: centre of pressure

Cambered airfoil: where is the centre of pressure at zero lift and as α increases? And in drag-free levelled flight?

MUST KNOW

- At the zero-lift angle α_0 the lift is zero but the moment is not, so the CP is at **infinity behind** the trailing edge.
- As α increases, the CP **comes in from behind and moves forward**, approaching the quarter chord $c/4$ (the aerodynamic centre) but never passing it.
- Drag-free levelled flight: lift equals weight and pitch equilibrium needs the lift line through the centre of gravity, so the CP is **at the centre of gravity**.

$$\frac{x_{cp}}{c} = \frac{1}{4} - \frac{c_{m,ac}}{c_l}$$

Here x_{cp} is the CP position from the leading edge, c the chord, $c_{m,ac} < 0$ the moment coefficient about the aerodynamic centre (almost constant) and c_l the lift coefficient, which is zero at α_0 and grows with α.

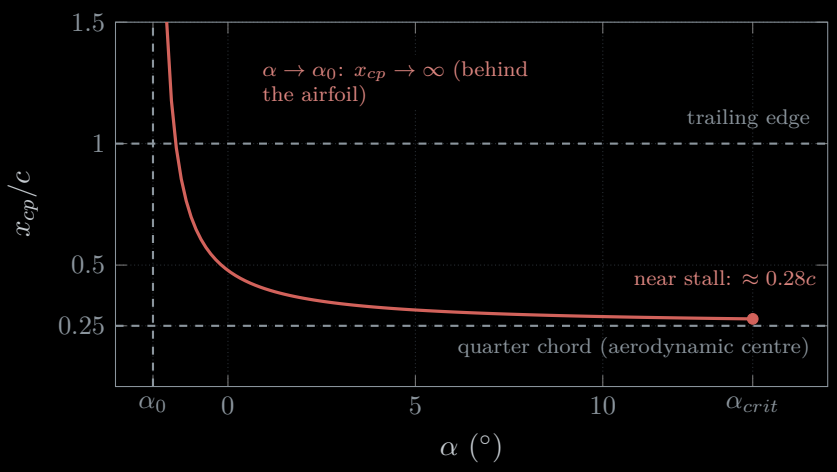


Figure 9.3. Centre of pressure against angle of attack for a cambered airfoil. The dashed line is the zero-lift angle.

HOW TO REMEMBER

A symmetric airfoil has $c_{m,ac} = 0$ and the CP stays at $c/4$. A cambered one has a nose-down moment even with no lift; dividing that fixed moment by a tiny lift puts the CP far away, and dividing it by a big lift brings the CP back to $c/4$.

1.4 Slide question 4: airfoil shape and the sound barrier

Which effect does the airfoil shape have on the sound barrier?

MUST KNOW

- Air accelerates over the upper surface, so **sonic flow appears locally before the flight speed is sonic**. M_{crit} is the flight Mach where $M = 1$ first appears on the airfoil; M_{DD} (drag divergence) is where the shocks separate the flow and drag rises sharply. Always $M_{crit} < M_{DD}$.
- A **thinner** airfoil accelerates the flow less: both Mach numbers move up.
- A **supercritical** airfoil (flat upper surface) gives a weaker shock, further aft: it raises M_{DD}.
- **Sweep** also delays the effect (only the velocity normal to the leading edge counts).
- Aircraft *can* fly above M_{crit}; they just stay below M_{DD}.

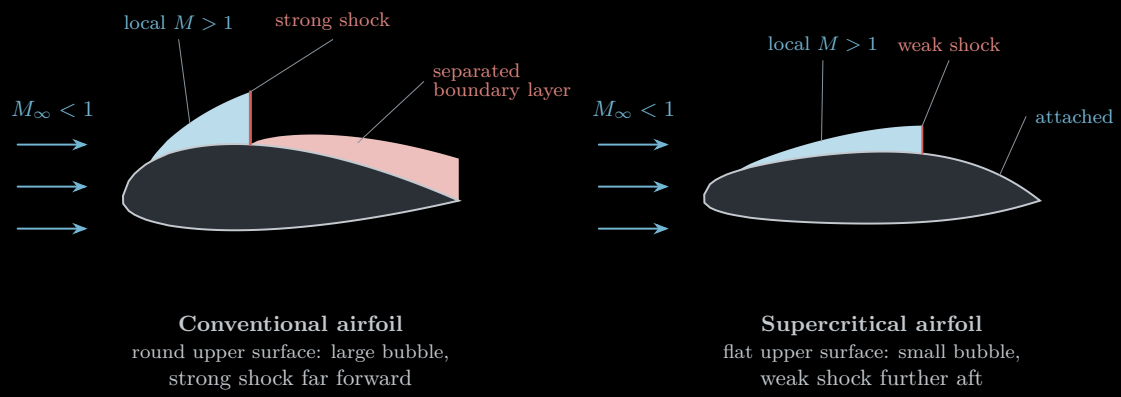


Figure 9.4. Conventional airfoil (strong shock, separation) against supercritical airfoil (weak shock, attached flow).

1.5 Slide question 5: aerodynamic efficiency curves

Shape of the efficiency curve for an airfoil (2D) and for an aircraft (3D).

MUST KNOW

Efficiency is c_l/c_d for the airfoil and $L/D = C_L/C_D$ for the aircraft, against α.

- **Zero** at the zero-lift angle, **rises** quickly (lift grows, drag barely), reaches a **maximum**, then **falls** near stall (separation: drag up, lift flat).
- The 3D curve is **much lower** than the 2D one: induced drag plus the parasitic drag of the other parts. Typical: transport 10–20, glider up to 70, hypersonic below 4.
- At the 3D maximum, parasitic and induced drag are **equal**.

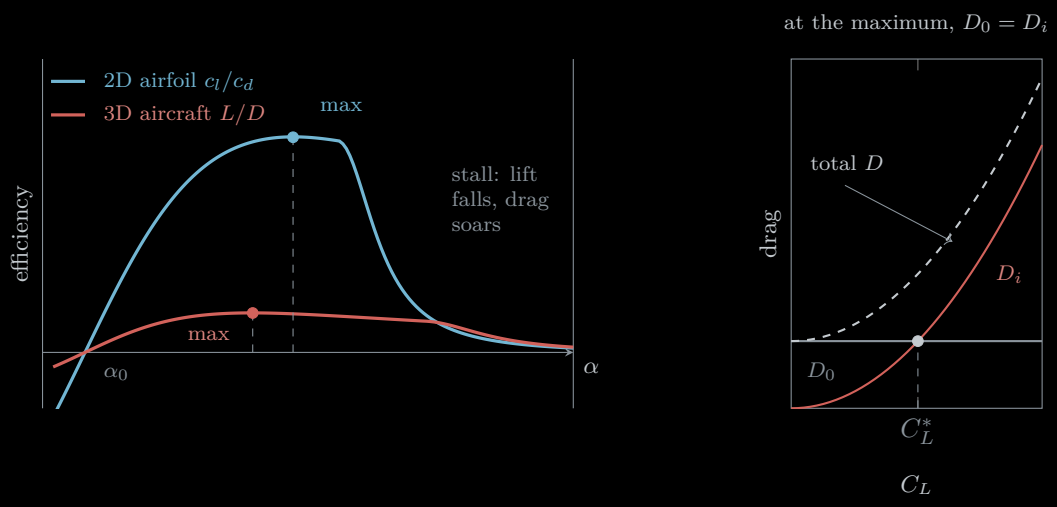


Figure 9.5. Efficiency against angle of attack, 2D and 3D (qualitative).

1.6 Slide question 6: friction coefficient against Reynolds number

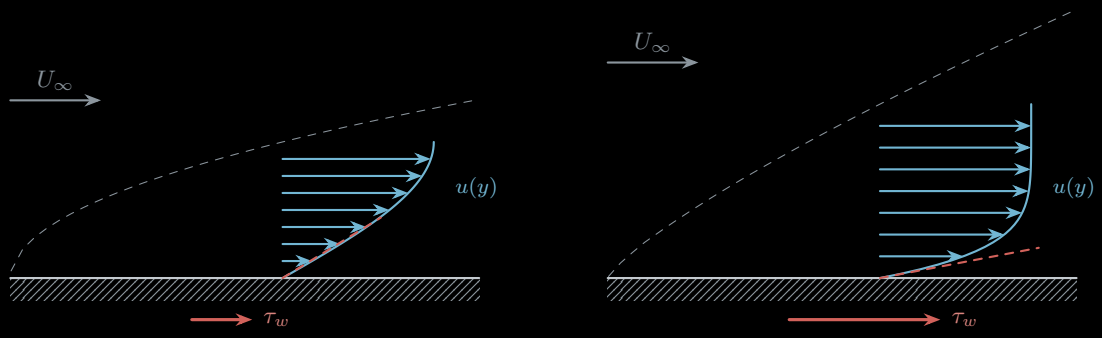
How does the C_f curve against Re look?

MUST KNOW

On a log–log plot:

- **Laminar**: a straight line falling as $C_f = 1.328/\sqrt{Re}$.
- **Turbulent**: also falls with Re, but more slowly, and **above** the laminar curve (fuller velocity profile, larger shear at the wall).
- **Roughness**: beyond the **cutoff Reynolds number** the turbulent curve **flattens** into a plateau. A rougher surface has a higher plateau and reaches it at lower Re.

Here C_f is the skin-friction coefficient (wall shear stress over $\frac{1}{2}\rho V^2$) and Re the Reynolds number ($\rho V x/\mu$).



Laminar
thick, rounded profile:
small slope at the wall

Turbulent
thinner near-wall layer, full profile:
steep slope at the wall

Figure 9.6. Laminar and turbulent profiles: the steeper wall gradient of the turbulent one means more friction.

1.7 Slide question 7: induced and total drag

How can the induced drag be reduced? How do the induced, parasitic and total drag coefficients vary with speed in cruise?

MUST KNOW

- **Origin**: wingtip vortices create downwash, which tilts the lift backwards. Only finite (3D) wings have it.
- **Reduce it**: larger aspect ratio, elliptic lift distribution (higher Oswald factor e), winglets, lower C_L.
- **With speed** (level flight, $C_L \propto 1/V^2$): $C_{D,i} \propto 1/V^4$ falls fast; $C_{D,0}$ is about constant; total C_D falls and tends to $C_{D,0}$. The *forces* are different: $D_0 \propto V^2$, $D_i \propto 1/V^2$.

$$C_{D,i} = \frac{C_L^2}{\pi\, AR\, e}$$

Here $C_{D,i}$ is the induced-drag coefficient, C_L the lift coefficient, AR the wing aspect ratio (span squared over area) and e the Oswald efficiency factor ($e \leq 1$).

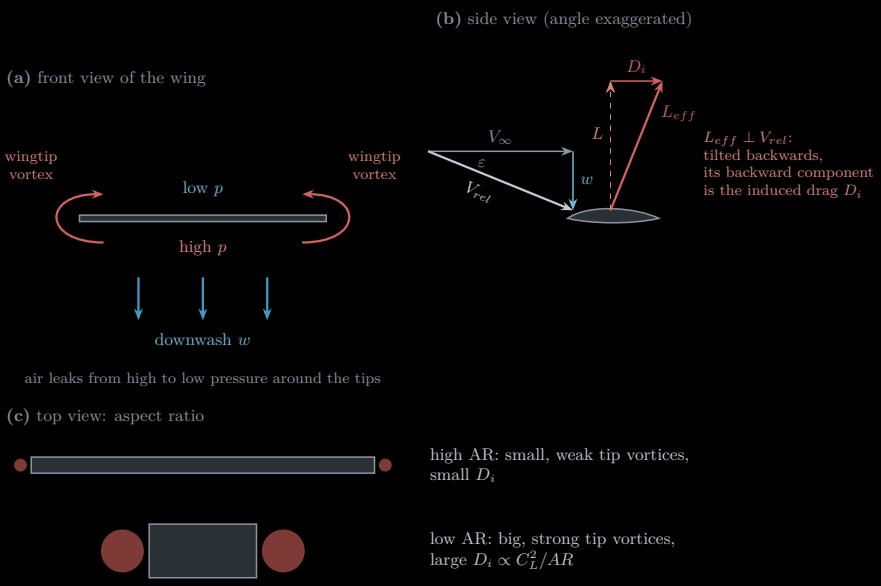


Figure 9.7. Tip vortices, downwash and the backward tilt of the lift that is the induced drag.

2 Essential answers to the professor's list of review questions

2.1 Question 1: ICAO, IATA, EASA, AESA and DGAC

Differences between ICAO, IATA, EASA, AESA and DGAC.

MUST KNOW

- **ICAO**: UN agency, worldwide; sets the international standards (SARPs) that states transpose. Does not inspect airlines.
- **IATA**: airlines' trade association, worldwide; commercial standards and codes; **no legal power**.
- **EASA**: European Union safety agency; common rules and certification (the FAA's counterpart).
- **AESA**: Spanish national authority; applies the rules, licenses, registers, inspects, sanctions.
- **DGAC**: Spanish ministry directorate for civil aviation policy and regulation.

Levels: global (ICAO, IATA), European (EASA), national (AESA, DGAC). **Rule makers**: ICAO, EASA, DGAC. **Enforcer in Spain**: AESA.

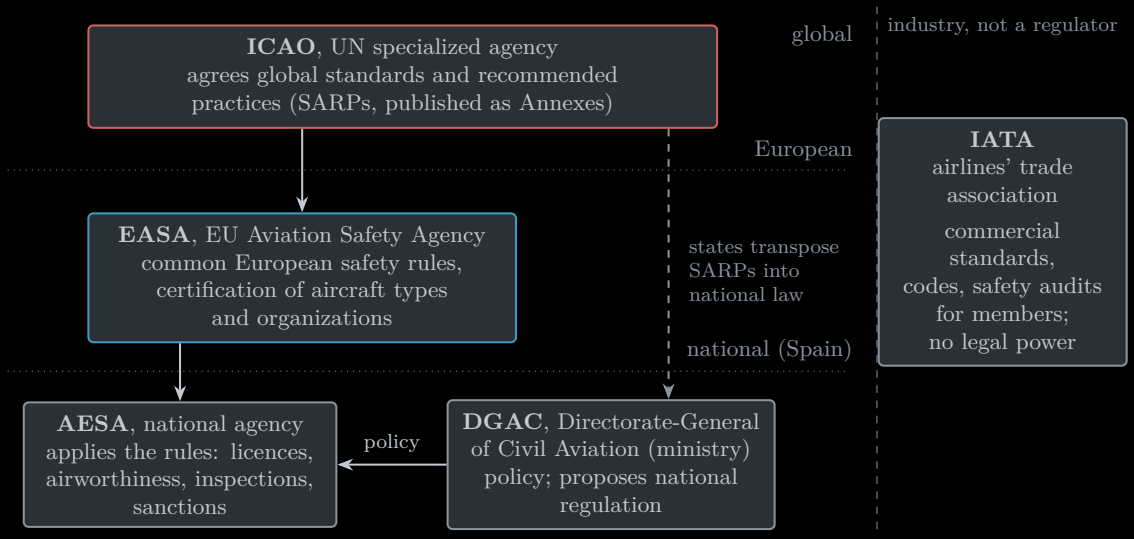


Figure 9.8. The five bodies by level; IATA is separate (airlines' association).

2.2 Question 2: navigation aids

SAME AS SLIDE QUESTION 1
Essential answer in Section 9.1.1 (p. 3).

2.3 Question 3: kinetic heating of A350, Concorde and SR-71

Temperature of the stagnant air on the skin of an A350, a Concorde and an SR-71.

MUST KNOW
The air touching the skin is brought to rest and its kinetic energy becomes internal energy (adiabatic compression), **not friction**. The stagnation temperature is

$$T_0 = T\,(1 + 0.2\,M^2)$$

Here T_0 is the stagnation temperature, T the static ambient temperature (ISA: 216.65 K from 11 to 20 km, 220.65 K at 24 km), M the flight Mach number and $0.2 = (\gamma - 1)/2$ with $\gamma = 1.4$.

	M	T (K)	T_0 (K)	T_0 (°C)
A350	0.85	216.65	248	−25
Concorde	2.02	216.65	393	120
SR-71	3.2	220.65	672	399

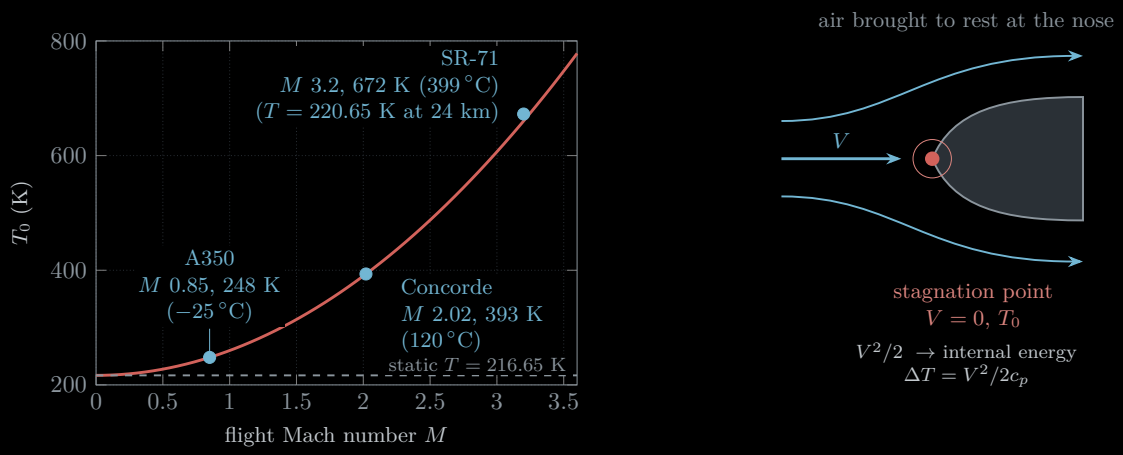


Figure 9.9. T_0 against Mach number, with the three aircraft, and the origin of the heating at the nose.

HOW TO REMEMBER
The factor $1 + 0.2M^2$ is 1.14 at $M = 0.85$, 1.8 at $M = 2$ and 2.8 at $M = 3.2$. Multiply about 217 K by it. The trend is the point: negligible for the airliner (aluminium), serious for Concorde (aluminium limit near Mach 2), extreme for the SR-71 (titanium).

2.4 Question 4: Breguet range equation for jets and propellers

Develop Breguet's equation for jets and propellers; when does it depend directly on speed?

MUST KNOW

$$R_{jet} = \frac{V}{g\,c_T}\,E\,\ln\frac{W_0}{W_f}, \qquad R_{prop} = \frac{\eta_p}{g\,c_P}\,E\,\ln\frac{W_0}{W_f}$$

Here R is the range, V the flight speed, g gravity, c_T the thrust-specific fuel consumption (kg/(N s)), c_P the brake-power-specific fuel consumption (kg/J), η_p the propeller efficiency, $E = L/D$ the aerodynamic efficiency and W_0, W_f the initial and final weights.

- **Jet: depends directly on V.** Fuel burned per second is proportional to the thrust (= drag), not to the speed, so flying faster covers more distance per kilogram.
- **Propeller: does not depend on V.** Fuel flow is proportional to the power $D\,V$, which grows with speed just as the distance per second does, so V cancels.
- Indirectly both depend on V through E, c_T and η_p.

HOW TO REBUILD IT IN FOUR LINES

(1) Level flight: $L = W$, $T = D = W/E$. (2) Fuel burn: $\mathrm{d}W = -g\,c_T\,T\,\mathrm{d}t$ for a jet, or $-g\,c_P\,P\,\mathrm{d}t$ with $P = DV/\eta_p$ for a propeller. (3) Distance: $\mathrm{d}x = V\,\mathrm{d}t$. (4) Divide, take constants out and integrate $\mathrm{d}W/W$, which gives the logarithm. The aircraft gets lighter, so the range is accumulated over the weight change.

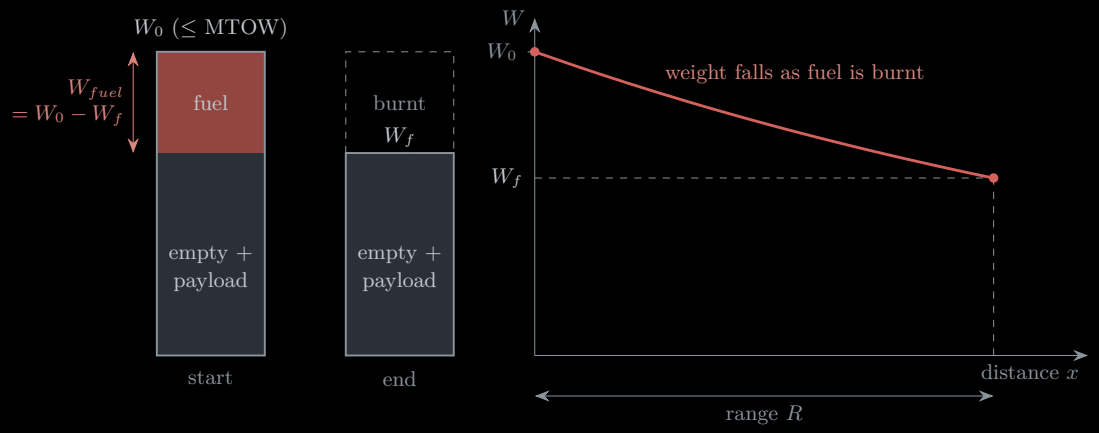


Figure 9.10. The weight falls from W_0 to W_f as fuel is burnt; the range is the distance covered meanwhile.

2.5 Question 5: specific fuel consumption data in SI units

Convert TSFC and BSFC data to SI.

MUST KNOW

$$1\ \frac{\text{lb}}{\text{lbf}\,\text{h}} = 2.83 \cdot 10^{-5}\ \frac{\text{kg}}{\text{N}\,\text{s}}, \qquad 1\ \frac{\text{lb}}{\text{hp}\,\text{h}} = 1.69 \cdot 10^{-7}\ \frac{\text{kg}}{\text{J}}$$

These come from 1 lb = 0.4536 kg, 1 lbf = 4.448 N, 1 hp = 745.7 W and 1 h = 3600 s. TSFC is fuel per unit of *thrust*; BSFC is fuel per unit of *shaft power*. They cannot be compared directly.

Engine (cruise)	Typical value	In SI
High-bypass turbofan	0.5–0.6 lb/(lbf h)	1.4–$1.7 \cdot 10^{-5}$ kg/(N s)
Turbojet, no afterburner	0.9–1.1 lb/(lbf h)	2.5–$3.1 \cdot 10^{-5}$ kg/(N s)
Turboprop	0.5–0.6 lb/(hp h)	0.85–$1.0 \cdot 10^{-7}$ kg/J
Piston engine	0.4–0.5 lb/(hp h)	0.68–$0.85 \cdot 10^{-7}$ kg/J

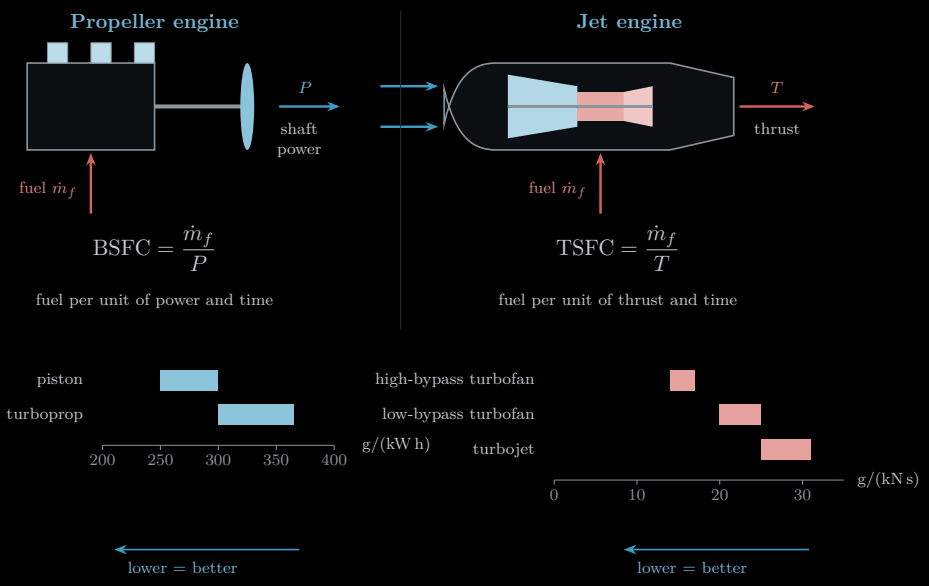


Figure 9.11. BSFC and TSFC as fuel burned per unit of useful output; lower is better.

2.6 Question 6: centre of pressure of a cambered airfoil

SAME AS SLIDE QUESTION 3
Essential answer in Section 9.1.3 (p. 5).

2.7 Question 7: skin-friction coefficient and roughness

C_f curves for laminar and turbulent flow; effect of roughness.

MUST KNOW

- **Laminar**: $C_f = 1.328/\sqrt{Re_x}$ (Blasius). Falls with Re. The layer turns turbulent near $Re_x \approx 5 \cdot 10^5$.
- **Turbulent, smooth**: higher than laminar at the same Re and falling more slowly, roughly $C_f \approx 0.074/Re_x^{0.2}$.
- **Roughness** (height k of the bumps): at low Re the viscous sublayer hides the bumps and the curve equals the smooth one; as Re grows the bumps poke out and C_f rises above it; at high Re the drag is pressure on the bumps and C_f becomes **constant** (plateau). **Rougher** (smaller x/k): higher plateau, reached sooner. The laminar regime is hardly affected.

Here $Re_x = \rho V x/\mu$ is the Reynolds number at distance x from the leading edge and μ the dynamic viscosity.

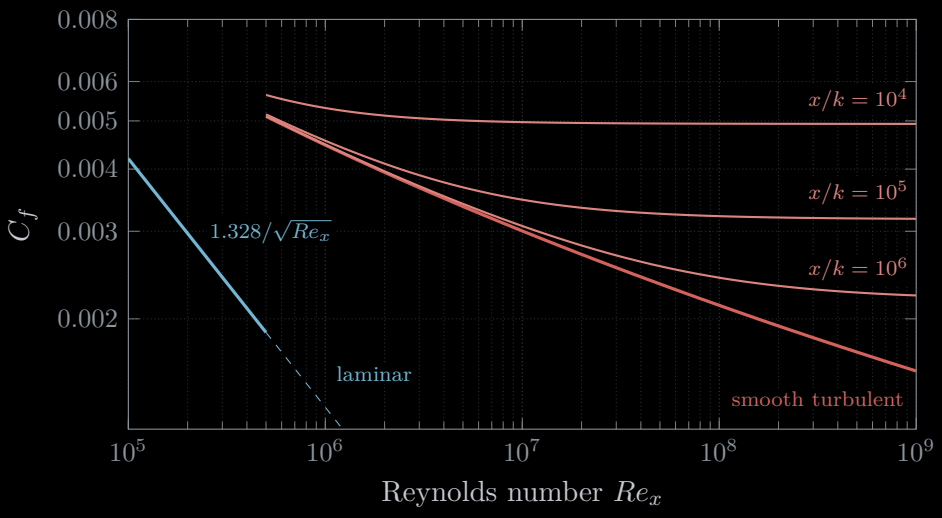


Figure 9.12. C_f against Re_x: laminar, smooth turbulent and rough turbulent curves that separate from the smooth one and flatten.

2.8 Question 8: aerodynamic efficiency, derivation and trends

Derive the aerodynamic efficiency, sketch it, and give the trends when induced drag is much larger or much smaller than parasitic drag.

MUST KNOW

With the parabolic polar $C_D = C_{D,0} + k\,C_L^2$:

$$E = \frac{C_L}{C_{D,0} + k\,C_L^2}, \qquad C_L^* = \sqrt{\frac{C_{D,0}}{k}}, \qquad E_{max} = \frac{1}{2\sqrt{k\,C_{D,0}}}, \qquad k = \frac{1}{\pi\,AR\,e}$$

Here E is the aerodynamic efficiency L/D, C_L and C_D the lift and drag coefficients, $C_{D,0}$ the parasitic (zero-lift) drag coefficient, k the induced-drag factor, AR the aspect ratio, e the Oswald factor and C_L^* the lift coefficient at the maximum.

- **Maximum** when $C_{D,0} = k\,C_L^2$: **parasitic and induced drag are equal**. It grows with larger AR and a cleaner aircraft.
- **Induced $\ll$ parasitic** (small C_L): $E \approx C_L/C_{D,0}$, a **straight line** rising with C_L.
- **Induced $\gg$ parasitic** (large C_L): $E \approx 1/(k\,C_L)$, a **hyperbola** falling with C_L.

HOW TO REBUILD IT

Derive $E(C_L)$ and set $\mathrm{d}E/\mathrm{d}C_L = 0$: the numerator becomes $C_{D,0} - k\,C_L^2 = 0$. Sketch: a line from the origin, a hyperbola at the right, and a flat-topped peak between them. At half or twice the optimum C_L the efficiency is still 80 % of the maximum.

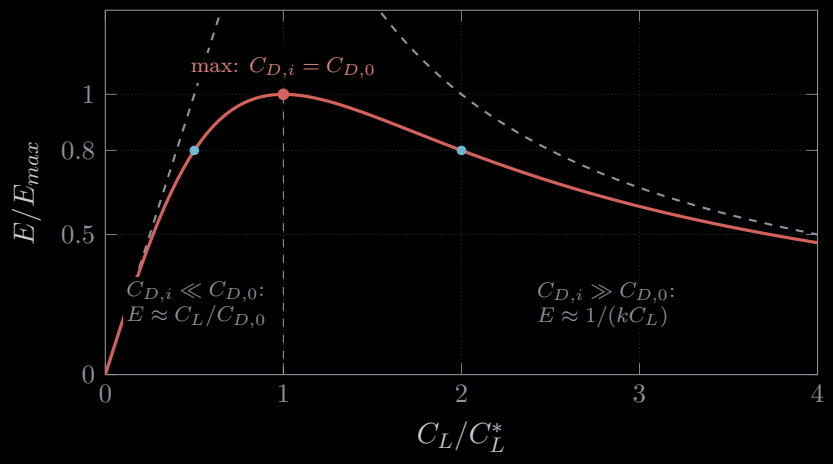


Figure 9.13. Efficiency against C_L with its two asymptotes (normalised by the optimum values).

2.9 Question 9: total drag in level flight and minimum-drag speed

Derive total drag and the minimum-drag speed; effect of altitude, aspect ratio, C_{D0} and wing loading.

MUST KNOW

In level flight $L = W$, so $C_L = 2W/(\rho V^2 S)$ and

$$D = \underbrace{\tfrac{1}{2}\rho V^2 S\, C_{D0}}_{\text{parasitic } \propto V^2} + \underbrace{\frac{2k\,W^2}{\rho V^2 S}}_{\text{induced } \propto 1/V^2}, \qquad V_{md} = \sqrt{\frac{2W}{\rho S}}\left(\frac{k}{C_{D0}}\right)^{1/4}, \qquad D_{min} = 2W\sqrt{k\,C_{D0}}$$

Here D is the drag, ρ the air density, V the true airspeed, S the wing area, W the weight, C_{D0} the zero-lift drag coefficient, $k = 1/(\pi\, AR\, e)$ the induced-drag factor, V_{md} the minimum-drag speed and D_{min} the minimum drag. At the minimum, **parasitic = induced drag**, and $D_{min} = W/E_{max}$.

Change	V_{md}	D_{min}
Higher altitude (lower ρ)	increases ($\propto \rho^{-1/2}$)	unchanged
Higher aspect ratio	decreases slightly	decreases
Higher C_{D0}	decreases slightly	increases
Higher wing loading W/S	increases ($\propto \sqrt{W/S}$)	increases if W grows

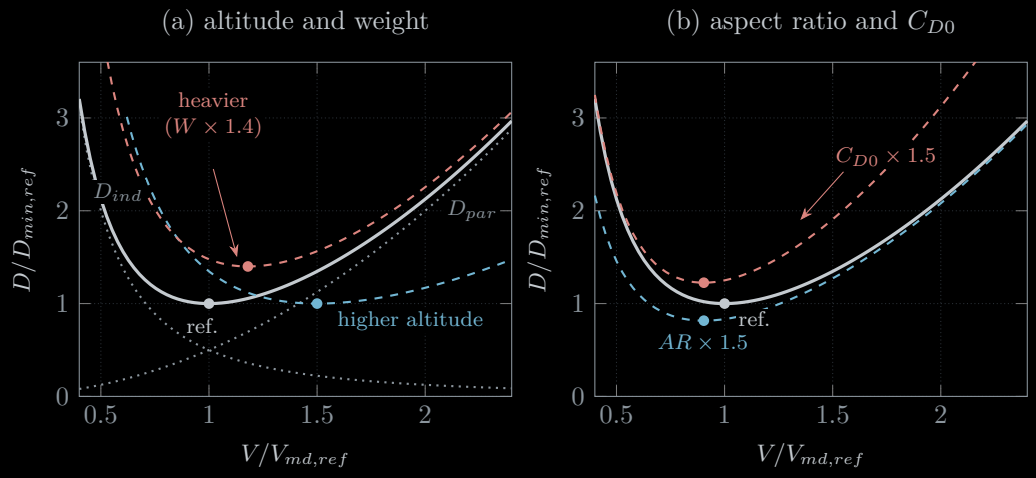


Figure 9.14. Total drag curve and how it shifts with altitude, weight, aspect ratio and C_{D0} (dots are the minima).

HOW TO REMEMBER

The parasitic branch ($\propto V^2$) wins at high speed, the induced branch ($\propto 1/V^2$) wins at low speed, and the minimum is where they cross. Altitude only stretches the speed axis (same D_{min}, higher true speed); a cleaner or longer wing lowers the whole curve.

2.10 Question 10: propulsive efficiency of air-breathing engines

Develop the propulsive efficiency; what does it depend on; effect of the inlet and outlet speeds.

MUST KNOW

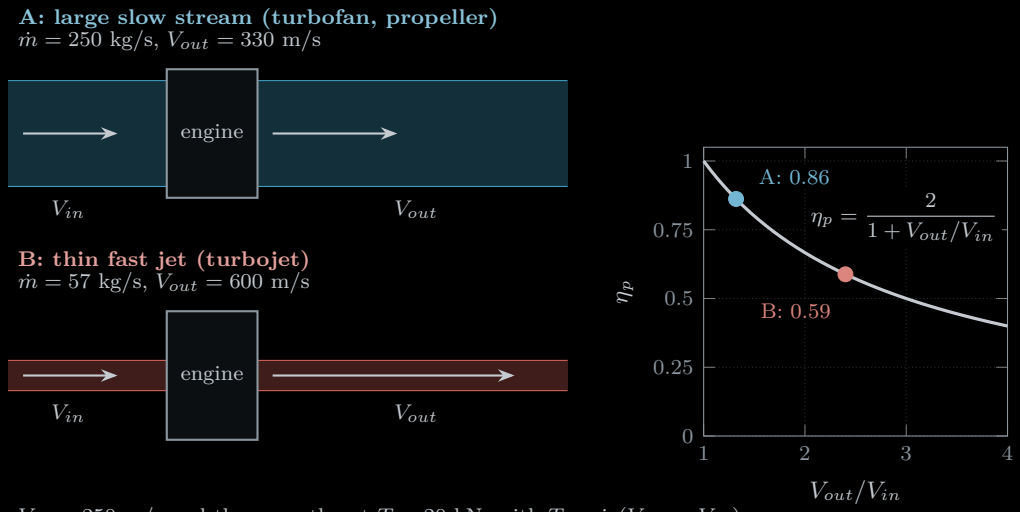
$$\eta_p = \frac{T\,V_0}{\frac{1}{2}\dot{m}\,(V_e^2 - V_0^2)} = \frac{2}{1 + V_e/V_0}$$

Here η_p is the propulsive efficiency, $T = \dot{m}\,(V_e - V_0)$ the thrust, $\dot{m}$ the air mass flow, V_0 the inlet speed (flight speed) and V_e the exit speed. It depends **only on** V_e/V_0.

- **Inlet speed V_0 higher**: η_p **increases** (tends to 1 as $V_0 \to V_e$), but the thrust falls to zero there. At $V_0 = 0$, $\eta_p = 0$.
- **Exit speed V_e higher**: η_p **decreases**, while the thrust increases.
- So for a given thrust it is better to accelerate a *large* mass of air *a little*: propellers and high-bypass fans.

HOW TO REBUILD IT

Useful power $= T\,V_0$; power given to the air = change in kinetic energy per second $= \frac{1}{2}\dot{m}\,(V_e^2 - V_0^2)$. Divide, and cancel $(V_e - V_0)$. The difference is the kinetic energy left in the wake, which is lost.



$V_{in} = 250$ m/s and the same thrust $T = 20$ kN, with $T = \dot{m}\,(V_{out} - V_{in})$.
Wake energy lost $= \frac{1}{2}\dot{m}\,(V_{out} - V_{in})^2$.

Figure 9.15. Same thrust: a wide slow stream wastes less kinetic energy in the wake than a thin fast jet.

2.11 Question 11: propeller selection (Lab session 2)

Complete the propeller selection exercise of Lab session 2.

MUST KNOW

The data of the lab are not in these notes, so only the method is given.

$$J = \frac{V}{nD}, \qquad C_P = \frac{P}{\rho n^3 D^5}, \qquad C_T = \frac{T}{\rho n^2 D^4}, \qquad \eta = \frac{J\,C_T}{C_P}$$

Here J is the advance ratio, V the flight speed, n the rotation rate (rev/s), D the diameter, C_P and C_T the power and thrust coefficients, P the shaft power, T the thrust, ρ the air density and η the propeller efficiency.

1. Fix the design point: P, V, ρ (cruise).
2. Choose D and n (keep the helical tip Mach $\sqrt{(\pi n D)^2 + V^2}/a$ below about 0.8).
3. Compute J and C_P.
4. Enter the propeller chart with (J, C_P) and read η and the blade pitch. Choose the propeller whose peak lies there.
5. Thrust: $T = \eta\, P/V$.

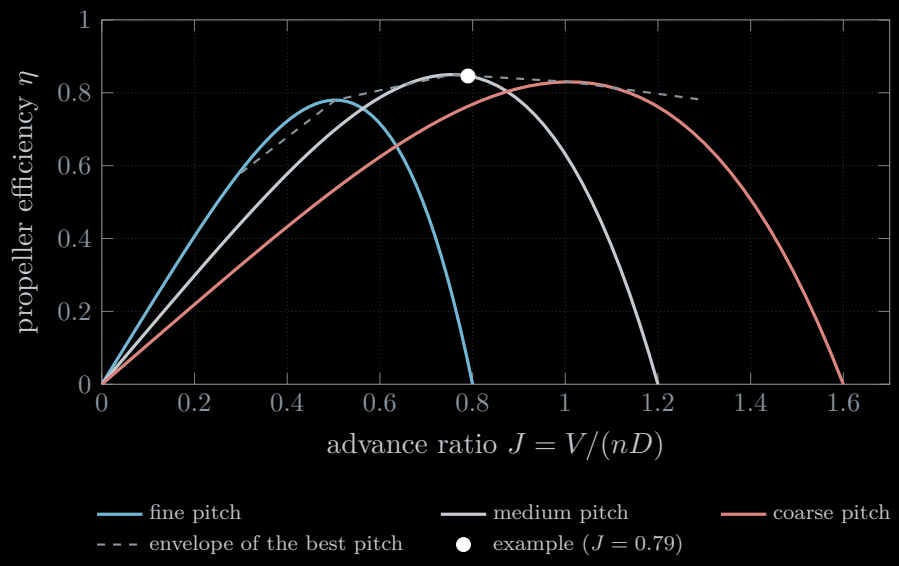


Figure 9.16. Efficiency against J for three blade pitches (schematic); the dashed envelope is the best pitch.

2.12 Question 12: specific impulse of chemical rockets

Typical specific impulse of chemical rocket engines.

MUST KNOW

$$I_{sp} \approx 250 \text{ to } 450 \text{ s}, \qquad v_e = I_{sp}\, g_0$$

Here I_{sp} is the specific impulse, v_e the effective exhaust speed (2.5 to 4.5 km/s) and $g_0 = 9.81$ m/s^2.

- Solid boosters: about 250–280 s. Kerosene/oxygen: about 300–350 s. Hydrogen/oxygen: up to about 450 s (vacuum).
- Air-breathing engines reach thousands of seconds because they **do not carry the oxidizer**.
- Vacuum values are higher than sea-level ones.

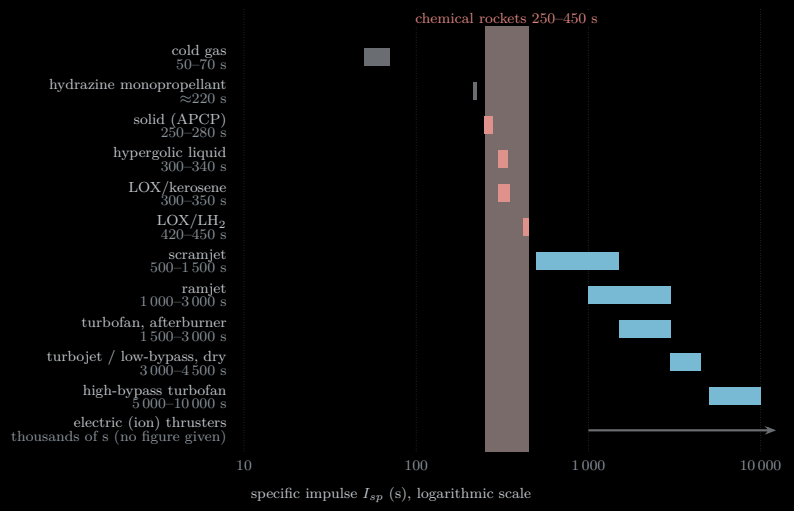


Figure 9.17. Typical specific impulse by propulsion type on a logarithmic scale; chemical rockets in the shaded band.

2.13 Question 13: mass estimate of SSTO and TSTO launchers

NOT IN TEST 1
Taught after unit 2.5. Its answer is in `Apuntes_Tecnologia_Aeroespacial_Anexo.pdf`.

2.14 Question 14: operating Mach ranges and specific impulse of air-breathing engines

Typical Mach ranges and specific impulses of turbofan, afterburning turbofan, ramjet and scramjet.

MUST KNOW

Engine	Mach	I_{sp} (s)
Turbofan	0–0.9	5 000–10 000
Turbofan with afterburner	up to 2–2.5	1 500–3 000
Ramjet	2–5 (up to 6)	1 000–3 000
Scramjet	5–15	500–1 500
Rocket (reference)	all	250–450

Ordering: below $M \approx 2$ the turbofan is the most economical, from $M \approx 3$ the ramjet, above $M \approx 6$ only the scramjet or the rocket. Orders of magnitude, not exact data. $I_{sp} = 1/(g\,\mathrm{TSFC})$, so $I_{sp} \approx 3600/\mathrm{TSFC}$ with TSFC in lb/(lbf h).

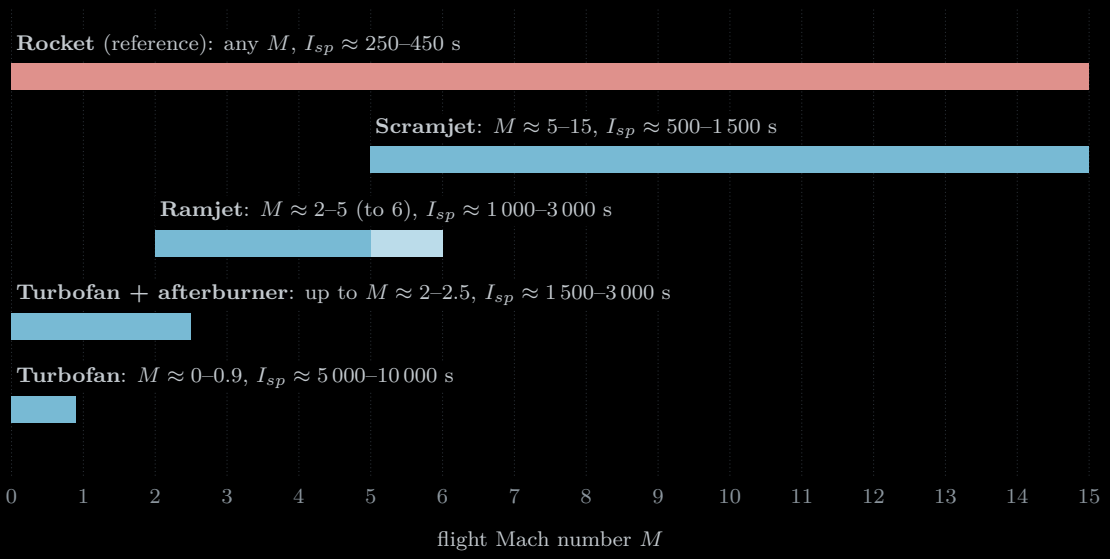


Figure 9.18. Mach window and specific impulse of each engine type.

2.15 Question 15: primary structure of a fuselage and of a wing

Elements of the semi-monocoque primary structure of a fuselage and of a wing, and their functions.

MUST KNOW
A thin **skin** stiffened by **lengthwise** members and by **transverse** members.

Element	Main function
Fuselage	
Skin	shape; carries shear, torsion and the cabin pressure
Stringers	axial loads of bending; stiffen the skin
Longerons	heavier lengthwise members; larger and concentrated loads
Frames	keep the cross-section shape; spread concentrated loads (wing, gear, tail)
Bulkheads	close the pressurised cabin; take large concentrated loads
Wing	
Skin	aerodynamic surface; with the spar webs forms the torsion box (torsion, shear)
Stringers	axial loads of wing bending; stiffen the skin
Spars	main beams: the web carries shear (lift), the flanges carry bending
Ribs	keep the airfoil shape; pass skin loads to the spars; support skin and stringers

Stringers, frames and ribs do not "prevent buckling": they cut the panel into narrower ones, which raises its buckling stress.

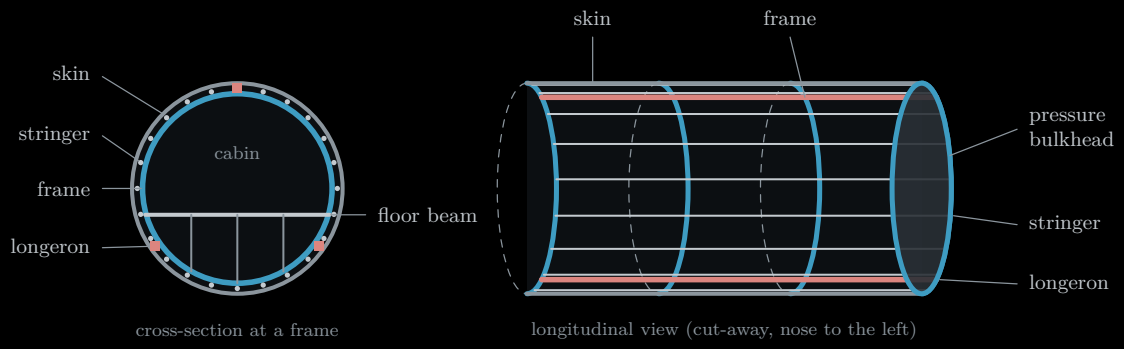


Figure 9.19. Semi-monocoque fuselage: skin, frames, stringers and longerons.

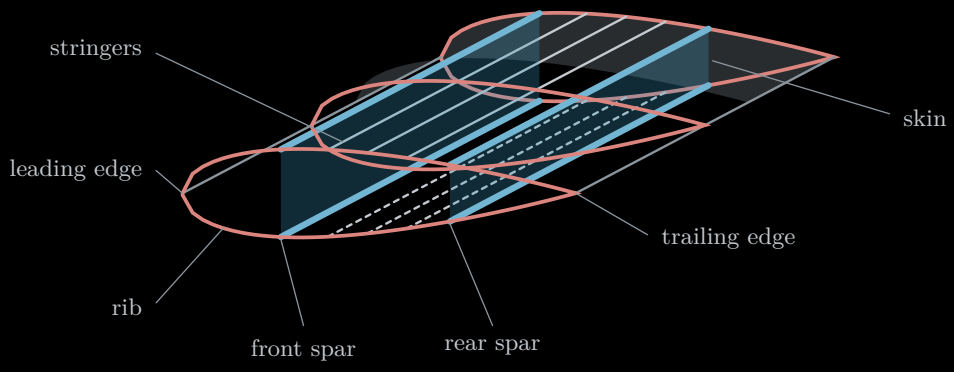


Figure 9.20. Semi-monocoque wing box: spars, ribs, stringers and skin.

2.16 Question 16: SSR, TCAS and ADS-B

NOT IN TEST 1
Taught after unit 2.5. Its answer is in `Apuntes_Tecnologia_Aeroespacial_Anexo.pdf`.

2.17 Question 17: specific range of a jet aircraft

Typical specific range curves against speed and altitude for passenger jets.

MUST KNOW

$$SR = \frac{V}{c\,D} = \frac{V\,(L/D)}{c\,W}$$

Here SR is the specific range (distance per unit of fuel weight), V the true airspeed, c the weight-specific fuel consumption, D the drag, L/D the aerodynamic efficiency and W the weight.

- **Against Mach** (fixed altitude): a flat maximum at a **best Mach**, a little above the minimum-drag speed ($\approx 1.32\,V_{md}$ with the parabolic polar) and limited by the wave-drag rise (about $M \approx 0.8$).
- **Against altitude**: it improves up to an **optimum altitude** and then falls (C_L too high, buffet).
- As fuel burns the aircraft is lighter and the optimum altitude **rises**: airliners do **step climbs**.

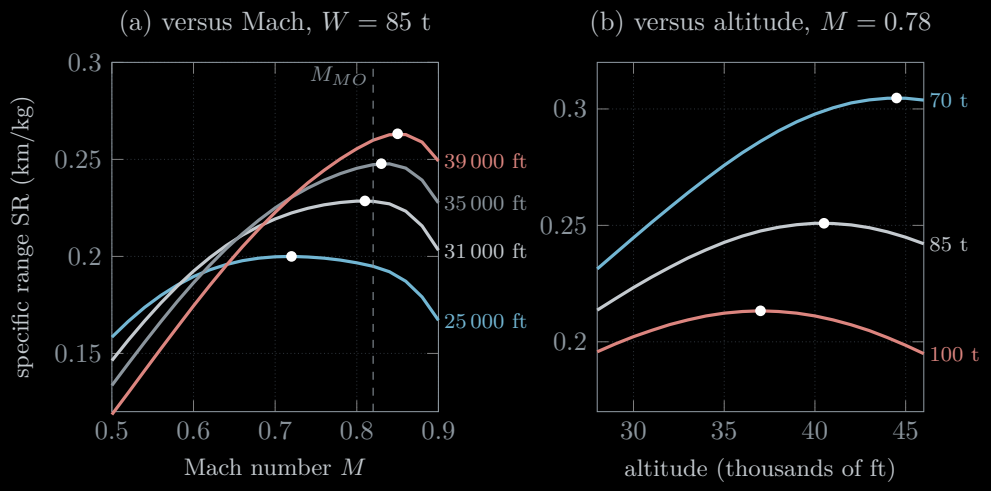


Figure 9.21. Specific range of a twin-jet airliner against Mach (several altitudes) and against altitude (several weights); dots mark the optima.

2.18 Question 18: vertical flight of helicopters

> **NOT IN TEST 1**
> Taught after unit 2.5. Its answer is in `Apuntes_Tecnologia_Aeroespacial_Anexo.pdf`.

2.19 Question 19: speed–altitude flight envelope of a passenger jet

Draw the speed–altitude envelope of a passenger jet; limits and causes. (The helicopter part is in the annex.)

> **MUST KNOW**
>
> - **Left: stall** ($C_L \leq C_{L,max}$), and at high altitude the **buffet** boundary. The stall speed is higher at altitude (lower ρ): $V_s \propto \sqrt{W/(\rho\, S\, C_{L,max})}$, with W the weight and S the wing area.
> - **Right:** V_{MO} (structure, flutter, dynamic pressure) at low altitude, then M_{MO} (wave drag, buffet, Mach tuck) at high altitude.
> - **Top: service ceiling**: thrust falls with density until there is no excess thrust left for climbing.
> - **Coffin corner**: the low-speed and high-speed buffet limits meet and the margin is almost zero.

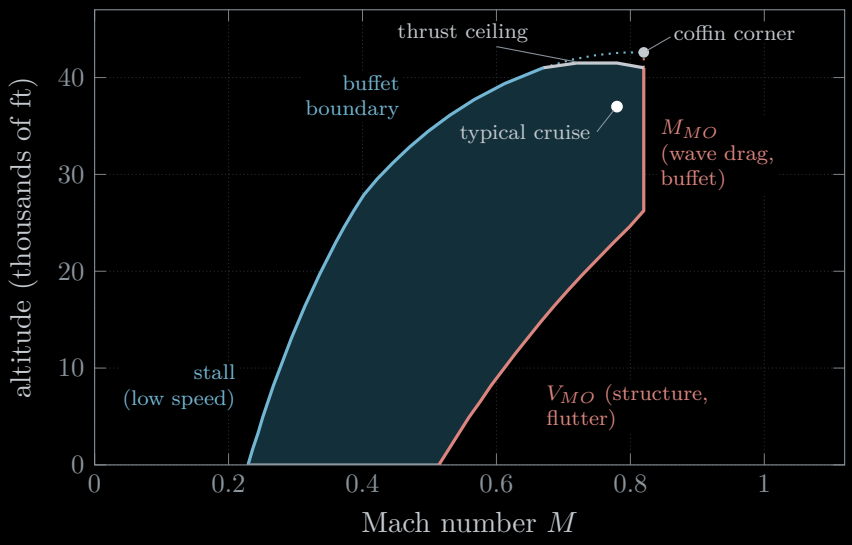


Figure 9.22. Speed–altitude envelope of a passenger jet; the shaded area is the permitted region.

2.20 Question 20: yaw control of a tandem twin-rotor helicopter

> **NOT IN TEST 1**
> Taught after unit 2.5. Its answer is in `Apuntes_Tecnologia_Aeroespacial_Anexo.pdf`.